

ECONOMETRIC METHODS FOR SOCIAL SPILLOVERS AND NETWORKS

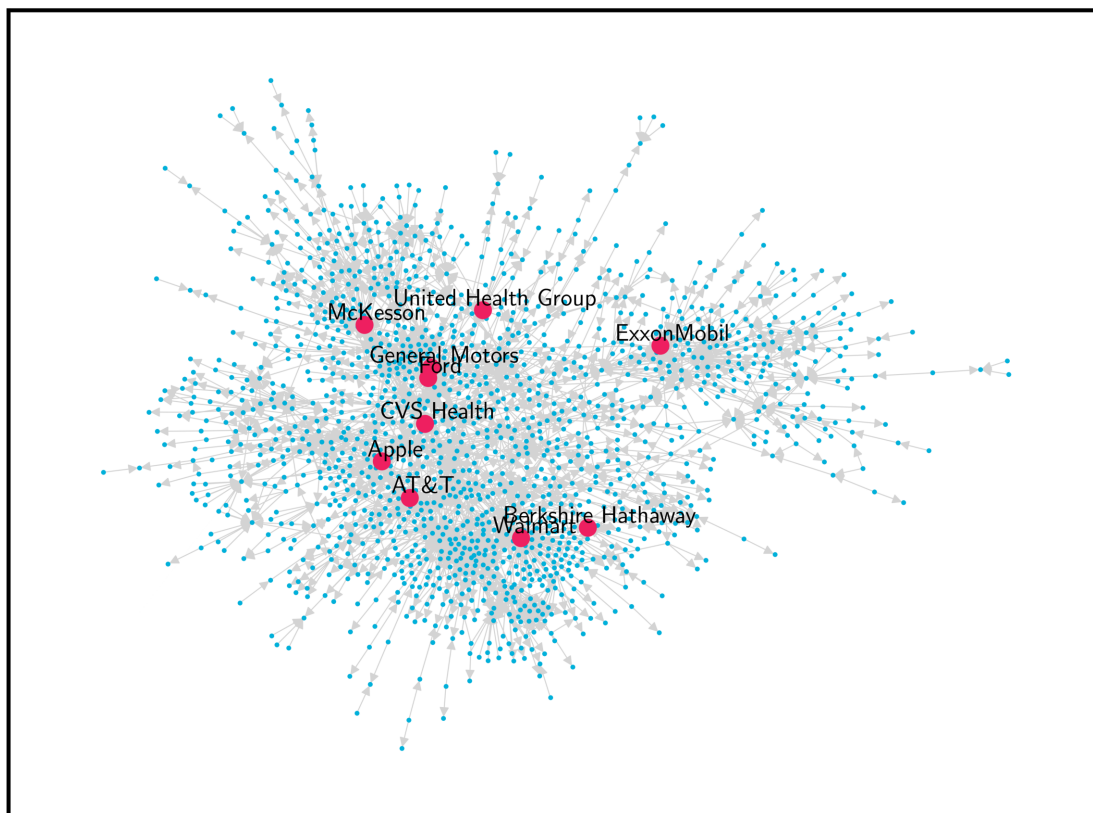
University of St. Gallen

September 21st to September 25th, 2026

“In a real sense all life is inter-related. All men are caught in an inescapable network of mutuality, tied in a single garment of destiny. Whatever affects one directly, affects all indirectly.” - Martin Luther King, *Letter from Birmingham Jail*

Course Description

This course will provide an overview of econometric methods appropriate for the analysis of social and economic networks. Many social and economic activities are embedded in networks. Furthermore, datasets with natural graph theoretic (i.e., network) structure are increasingly available to researchers. We will focus on (i) descriptive analysis of network data, (ii) dyadic regression methods, (iii) econometrics models of network formation admitting agent-specific heterogeneity and/or strategic interaction, and (iv) econometrics models of social interaction or peer group effects.



COURSE LOGISTICS

Instructor: Bryan Graham, Department of Economics, University of California - Berkeley

Email: bgraham@econ.berkeley.edu

Time: Monday 9/21 (8:15AM to 4PM), Tuesday 9/22 (8:15AM to 4PM), Wednesday 9/23 (8:15AM to 1:15PM), Thursday 9/24 (No Class), Friday 9/25 (8:15AM to 4PM); official hours will also be scheduled.

Notes: *We will break for lunch from noon to 2PM on Monday, Tuesday and Friday. We will likely end at 12:15PM on Wednesday so that we can attend the Brown Bag seminar. There is no class on Thursday, but I will hold office hours that day (likely in the morning).*

Prerequisites: The equivalent of a first year Ph.D. level sequence in econometrics. Specifically an understanding of probability and statistical inference at the level of Casella and Berger (1990, *Statistical Inference*), linear regression analysis at the level of Goldberger (1991, *A Course in Econometrics*) and some exposure to non-linear models (e.g., maximum likelihood, M-estimation, GMM). I will also assume a basic knowledge of applied linear/matrix algebra. Exposure to the theory of U-Statistics is useful, but not required.

Textbook: Readings preceded by a [r] in the course outline are “required” (i.e., should ideally be read prior to class), while those preceded by a [b] are for “background” (i.e., may be useful for students interested in additional material or empirical applications). Students may find the book by Graham & de Paula (2020a) useful (available for purchase [here](#)). We will also make extensive use of my recent *Handbook of Econometrics* chapter (Graham, 2020b). This chapter is available on ScienceDirect [here](#). Students who anticipate doing research involving networks should consider purchasing the textbooks by Jackson (2008), Newman (2010) and/or Goyal (2023), but doing so is not necessary.

GitHub: Supplemental course materials, including slides, lecture notes and computer programs, will be made available on GitHub at https://github.com/bryangraham/short_courses.

Computation: The bulk of class will be devoted to the formal development of the material, albeit with empirical illustrations as well as ample discussions of the various practicalities of implementation. However I do intend to reserve some class time for actual practice with computation. Computational examples will be done using Python. Python is a widely used general purpose programming language with good functionality for scientific computing. For those wishing to manage a Python environment on their personal computer, the Anaconda distribution, which is available for download at <https://www.anaconda.com/distribution/>, is a convenient way to get started. Some basic tutorials on installing and using Python, with a focus on economic applications, can be found online at <http://quant-econ.net>. Good books for learning Python, with some coverage of statistical applications, are Guttag (2013), VanderPlas (2017), and McKinney (2017). The code I will provide will execute properly in Python 3.9, which is (close to) the latest Python release.

Grades: Grades will reflect a combination of in class participation and performance on a final take-home assessment. Details of the final assessment will be provided in class.

COURSE OUTLINE (TENTATIVE/SUBJECT TO REVISION)

DATE	TOPIC	READINGS
9/21 (Monday) (8:15AM to 4PM)	DESCRIBING NETWORKS <i>Examples of networks</i> <i>Small worlds</i> <i>Degree distributions</i> <i>Homophily</i> <i>Triads</i>	[r] Jackson et al. (2017) [b] Atalay et al. (2011); Mizuno et al. (2014) [b] Apicella et al. (2012); Marotta et al. (2015) [b] Goyal et al. (2016); Seongjoo (2019) [b] Hoshino (2022) [b] Milgram (1967) [b] Mitzenmacher (2004) [b] McPherson et al. (2001) [b] Granovetter (1973); Jackson et al. (2012) [b] Holland & Leinhardt (1976)
	CENTRALITY, SHOCKS & DIFFUSION <i>Social learning</i> <i>Empirical studies</i> <i>Production networks</i>	[r] Graham & de Paula (2020b) [b] Galeotti et al. (2020); Banerjee et al. (2021) [b] Golub & Jackson (2010, 2012) [b] Banerjee et al. (2013) [b] Kim et al. (2015); Beaman et al. (2021) [b] Jackson & Zenou (2015); König et al. (2019) [r] Carvalho & Tahbaz-Salehi (2019); Carvalho et al. (2020) [b] Acemoglu et al. (2012, 2016)
9/22 (Tuesday) (8:15AM to 4PM)	SUBGRAPHS & DYADIC REGRESSION	[r] Graham (2020b, Sections 2 & 7) [b] Bickel et al. (2011); Bhattacharya & Bickel (2015) [r] Graham (2020a), Menzel (2021) [b] Fafchamps & Gubert (2007) [b] Rose (2004); Santos Silva & Tenreiro (2006) [r] Graham (2017), [b] Chatterjee et al. (2011)
9/23 (Wednesday) (8:15AM to 1:15PM)	STRATEGIC INTERACTION:	[r] de Paula (2013, 2020) [r] Graham (2020b, Section 8) [r] Graham & Pelican (2020); Pelican & Graham (2019) [r] Blitzstein & Diaconis (2011); McDonald et al. (2007) [b] Miyauchi (2016); de Paula et al. (2018) [r] Christakis et al. (2020) [r] Graham & Pelican (2023)
9/25 (Friday) (8:15AM to 4PM)	PEER AND NEIGHBORHOOD EFFECTS	[r] Manski (1993), Graham (2008) [r] Graham (2018), Bramoullé et al. (2009); Bramoullé et al. (2014) [b] Angrist (2014)

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